



DEPI Data Analysis Team 5 Project

ClimateScope

*An End-to-End Interactive Climate Change &
Environmental Sustainability Data Analysis Platform*

Project Team:

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1. Project Overview & Concept

1.1 Project Idea:

ClimateScope is an end-to-end interactive data analysis and visualization platform designed to monitor, track, and analyze the critical indicators of climate change and environmental sustainability across global regions over multi-decade time horizons. Developed under the Digital Egypt Pioneers Initiative (DEPI) Data Analysis Track, the project synthesizes extensive multidimensional datasets including historical global temperatures, greenhouse gas emissions, water resource volumes, natural disaster impacts, and agricultural nutrients into a unified, actionable analytical experience. The platform delivers insights through two distinct interactive deployment channels: a sophisticated multi-page Power BI dashboard designed for dense exploratory filtering, and a reactive Streamlit application engineered for real-time web-based reporting.

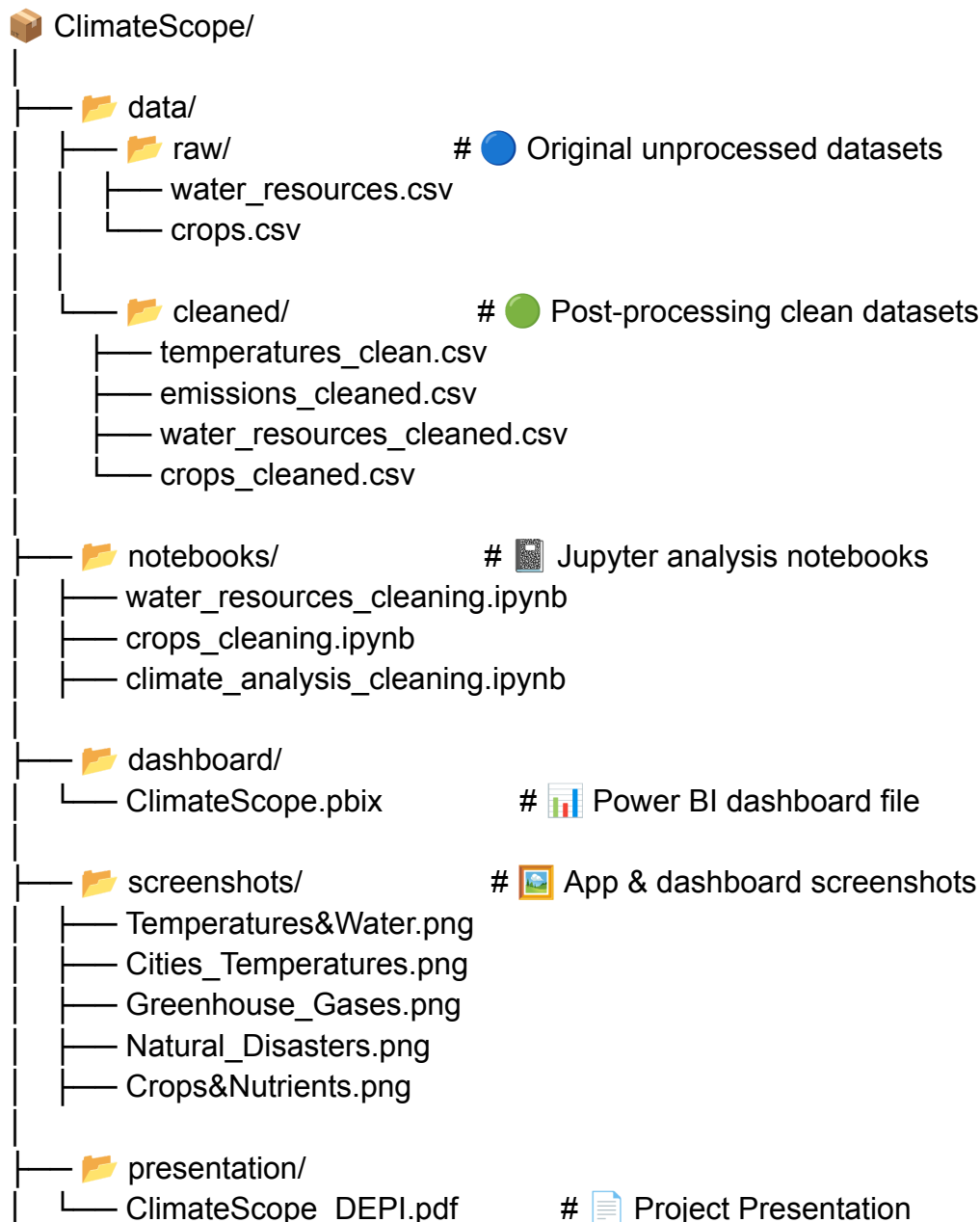
1.2 The Problem Statement:

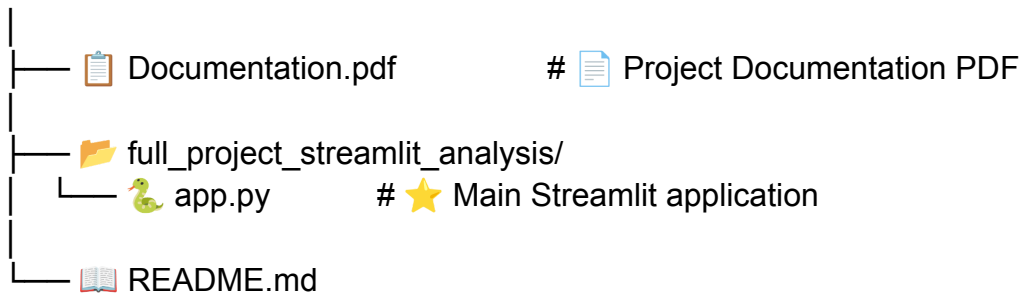
Modern climate research and environmental policy-making suffer from fragmented information silos. Crucial parameters such as atmospheric emission metrics, regional warming rates, hydrologic stress, and crop yield dependencies are often captured in isolated databases with varying formats and significant data quality issues (e.g., missing records, structural inconsistencies). This fragmentation prevents data analysts and policy stakeholders from recognizing multi-indicator correlations, such as how localized temperature anomalies intensify regional water scarcity and subsequently alter the N-P-K nutrient balance necessary for agricultural security. ClimateScope directly addresses this challenge by cleaning, consolidating, modeling, and visualizing these disparate environmental components into a cohesive star-schema data architecture, turning raw climate metrics into centralized, clear, and systemic insights.

2. System Architecture & Project Parts

2.1 Project Directory Structure

The ClimateScope platform maintains a modular engineering layout separating data stages, analytical notebooks, documentation, and operational web application scripts:





2.2 Technology Stack and Tools Used

- Data Extraction & Processing: Python, Pandas, NumPy
- Exploratory Data Analysis & Statistical Visualization: Matplotlib, Seaborn
- Data Modeling & Interactive Reporting: Power BI Desktop (DAX formulas, Slicers, Star Schema)
- Web Application Deployment: Streamlit Framework (Python)

2.3 Data Modeling & Database Structure

The project utilizes a relational Star Schema architecture optimized for high-performance time-series exploration and cross-indicator aggregations. The structure establishes explicit unidirectional 1-to-many (1:*) historical links from a centralized fact table to peripheral environmental dimensions:

- **Central Fact Hub:** Global Temperatures table, containing core historical chronological sequences (from 1900 to present day). Keys are mapped via date configurations (dt or year).
- **Connected Dimensions:**
 - Climate Change & Economic Data tables (capturing development indicators).
 - Water Resources table (storing per capita availability and stress trends).
 - Geographical Temperature Dimensions: State Temperatures, City Temperatures, Regional Temperatures, and Country Temperatures tables for granular spatial breakdowns.
- **Standalone Table:** Crop Data table remains disconnected to house isolated agricultural parameters (e.g., localized soil humidity and potassium levels) for standalone correlation inquiries.

3. Data Processing & ETL Pipeline

Data was thoroughly processed using a two-pronged Python pipeline implemented across automated notebook workflows:

Workflow A: Emissions & Climate Indicators Pipeline

1. **Library Selection & Ingestion:** Core analytics suites (Pandas, NumPy, Matplotlib, Seaborn) were imported, and normalized emissions datasets were loaded via CSV.
2. **Automated Imputation:** Built an automated processing loop tracking all columns. Numerical attributes (float64/int64) automatically filled missing rows via column mean metrics, while categorical attributes were imputed using the column mode.
3. **Data Type Optimization:** Forced Year and Value columns into standard numeric forms using errors='coerce' to capture anomalies, while trailing and leading spaces were stripped across all column headings.
4. **Analytical Subsetting:** Extracted top-10 emitters by identifying the maximum historical dataset year and charting emission metrics via a horizontal Seaborn viridis visual.
5. **Trend Tracking & Quantile Binning:** Aggregated areas to sum total volume outputs, isolated top-5 nations, and tracked growth via line plots. Segmented absolute emission values using a quantile-based scale (pd.qcut) into contribution categories: Bronze, Silver, Gold, and Platinum.
6. **Correlation Matrix Heatmapping:** Developed a pivot matrix indexed by year with distinct gas components to compute Pearson correlation coefficients, mapped using an annotated RdYlGn centered colormap.
7. **Historical Emission Gap Analysis:** Filtered records to map the earliest available historical year against the latest year for top emitters, charting changes using a clustered magma visualization. Data exported to Climate_Change_Final_Analysis.csv.

Workflow B: Water Resources Pipeline

1. **Ingestion & Standardization:** Standardized column layouts into clear variables: country, code, year, and water_value.
2. **Type Casting & Integrity Validation:** casted year to integer and water_value to numeric format. Removed duplicate records, ordered indices chronologically by country/year, and reset the dataframe index.
3. **Feature Engineering & Environmental Thresholds:** Created an environmental category named scarcity_level using np.select to map per-capita data into international classifications: Absolute Scarcity, Scarcity, Stress, and Abundant.
4. **Statistical Outlier Detection:** Applied Interquartile Range (IQR) analysis, defining Q1 (25th) and Q3 (75th) boundaries to flag values falling outside $1.5 \times$ IQR. Ranked countries inside individual historical blocks using dense ranking approaches.
5. **Decade-Block Aggregations:** Transformed year attributes via floor division to map data points into structured 10-year decade blocks (period). Results were exported directly into water_resources_final_analysis.csv.

Workflow c: Crop Recommendation & Nutrient Pipeline

1. **Investigation & Integrity Checking:** Conducted comprehensive data profiling on the crops dataset. Executed automated checks for missing parameters which confirmed 0 null entries across all structural bounds. Evaluated row duplication criteria, reporting 0 duplicate records to verify foundational consistency.
2. **Advanced Feature Engineering:**
 - **Total_Nutrients:** Synthesized soil vital signs by calculating the horizontal sum of Nitrogen (N), Phosphorus (P), and Potassium (K) values.
 - **Nutrient Ratios:** Formulated three separate stoichiometric metrics—N_P_Ratio, N_K_Ratio, and P_K_Ratio—utilizing a microscopic numerical epsilon value (ϵ) to avoid math exceptions or division-by-zero bounds during execution.
 - **Moisture_Index:** Generated a climate-crop moisture indicator by computing the direct dividend of rainfall to temperature records.
 - **Soil pH Categorization:** Constructed a categorical field named ph_category using logical conditional routing to segment crop environments into three distinct

agronomical tiers: Acidic ($\text{pH} < 6.0$), Neutral ($6.0 \leq \text{pH} \leq 7.5$), and Alkaline ($\text{pH} > 7.5$).

- **Temperature-Humidity Index (THI):** Derived bioclimatic environmental metrics by calculating the standardized heat index representation using the formula: $\text{ext}\{\text{temperature}\} + (0.55 \text{ imesext}\{\text{humidity}\})\$$.
- **Nutrient-to-Water Ratio:** Engineered a diagnostic indicator tracking the concentration of soil nutrients relative to precipitation by dividing Total_Nutrients by rainfall metrics (with stabilizing epsilon thresholds).

3. *Exploratory Data Analysis (EDA) & Statistical Graphics:*

- **Numerical Column Distributions:** Programmed a high-density 4×4 subplot visualization grid utilizing `sns.histplot` to visually capture and inspect frequency bounds and mathematical skewness across all continuous numeric indicators.
- **Correlation Heatmap Profiling:** Built an annotated statistical correlation heatmap across all numerical columns. The analysis isolated a high Pearson correlation index of 0.94 linking atmospheric humidity to the engineered THI, demonstrating humidity's massive baseline impact over regional thermal stress metrics.
- **Outlier Identification & Nutrient Distribution:** Profiled statistical distribution boundaries and extracted mean primary nutrition needs (N, P, K) segmented per crop label.
- **Nutrient Requirements by Crop (Top 5):** Isolated the top 5 distinct vegetation categories demanding the highest standalone averages for individual soil nutrients, mapping the variances using Matplotlib horizontal bar series charts for dashboard visualization inputs.

4. Dashboard Reports & Core Functions

The analytical platform exposes its primary system functions across 5 highly specialized interactive reports built in Power BI and mirrored via Streamlit components:

1) **Climate Overview Dashboard:**

Provides a sweeping overview of global climate indicators covering 193 nations. Core metrics show a global average temperature of 15.32°C, a high average humidity level of 71.48%, and an average water value of 24.02K. It includes a Nutrients-to-Water ratio of 1.98% indicating subtle ecosystem imbalances. Major

thermal hotspots identified include Umm Durman, Madras, Jiddah, Bangkok, and Ho Chi Minh City.

2) Greenhouse Gases Dashboard:

Tracks specific environmental impacts of gas concentrations. Quantifies CO₂ as the major global driver with an average index value of 76.84. Tracks Methane (CH₄) metrics, showing a gradual, long-term decline due to improved regional management. Nitrous Oxide (N₂O) levels exhibit stable, flat baselines over time. Line plots map sharp industrial acceleration spikes against global warming markers.

3) Water Resources & Temperature Dashboard:

Maps the inverse geographical correlations between rising thermal stress and hydrologic depletion. Highlights extreme water stress regions like Saudi Arabia, Djibouti, Mali, and Burkina Faso. It contrasts these against vast freshwater reserves in Iceland, Guyana, and Suriname. The dashboard tracks a continuous decline in global per capita water volumes over time.

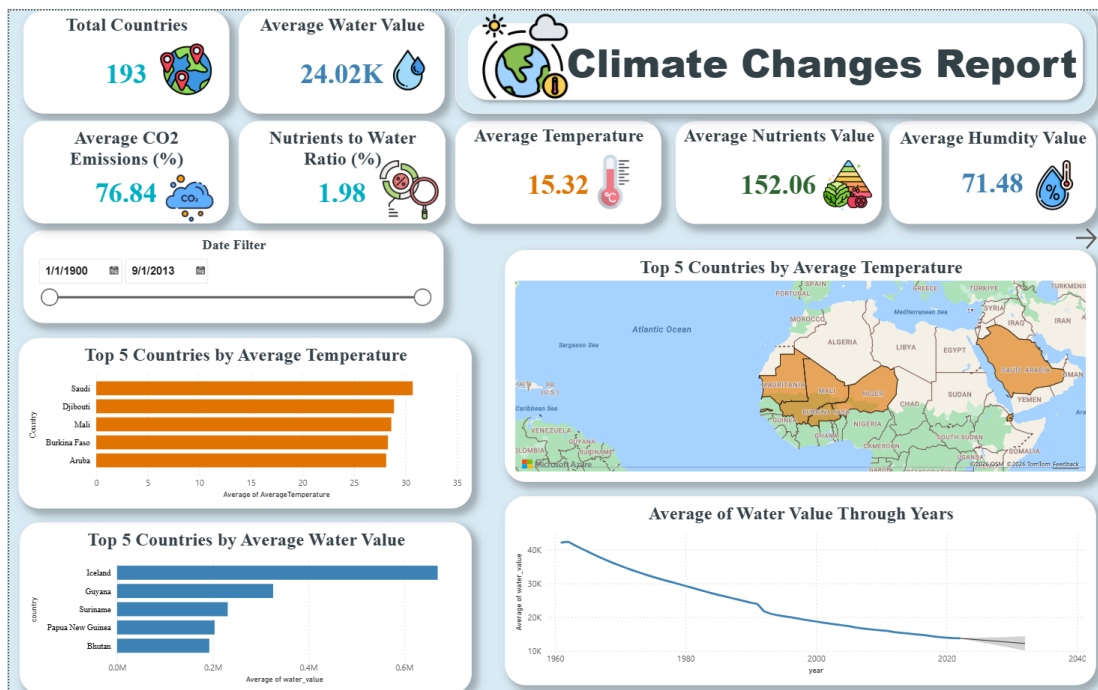
4) Natural Disasters Dashboard:

Analyzes spatial and temporal patterns across approximately 12,000 recorded catastrophic events. Identifies earthquakes and severe storms as leading drivers of global disaster fatalities, followed by floods and epidemics. Identifies Asia as the most vulnerable zone in human impact, with major concentrations across India, China, Indonesia, Myanmar, and Haiti.

5) Crops & Nutrients Dashboard:

Explores approximately 2,000 agricultural data logs to assess food security trends. Captures an average nutrient index value of 152.06 and an average N:P (Nitrogen to Phosphorus) ratio of 1.70. Models a positive relationship between humidity and nutrient metrics, showing peaks at 80-85% humidity. Identifies Muskmelon, watermelon, coffee, coconut, and cotton as crops with the highest N:P demand.

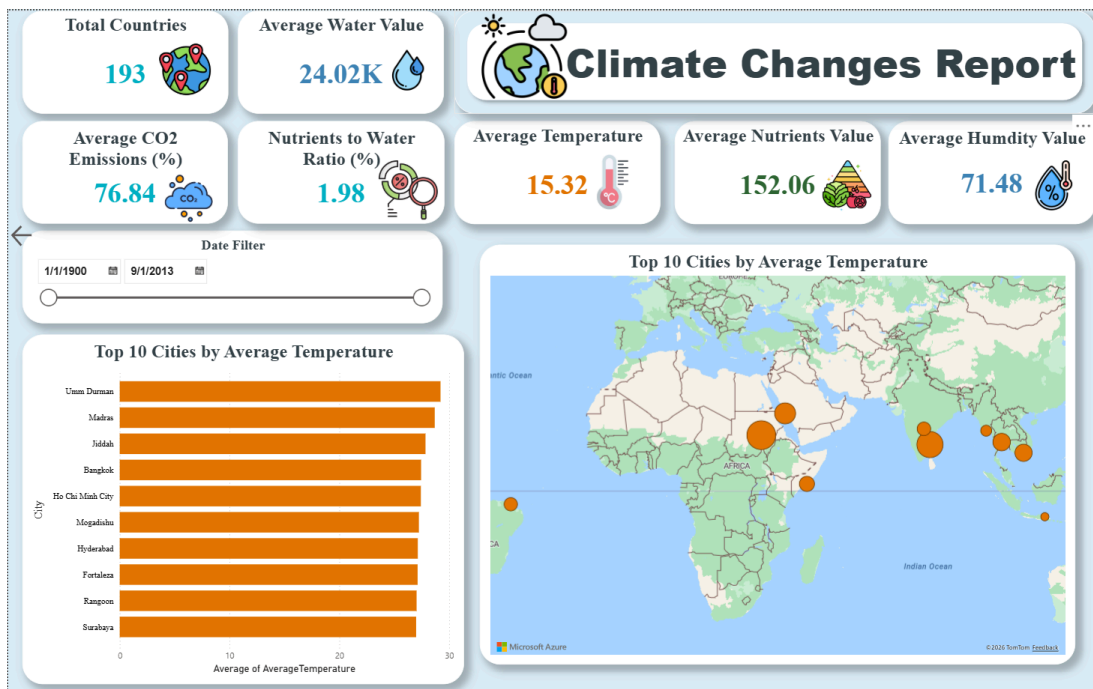
5. System Interface & Screenshots Manual



This section provides detailed breakdowns for each interactive dashboard screen, specifying operational user filters and technical features:

Screen 1: Climate & Water Overview

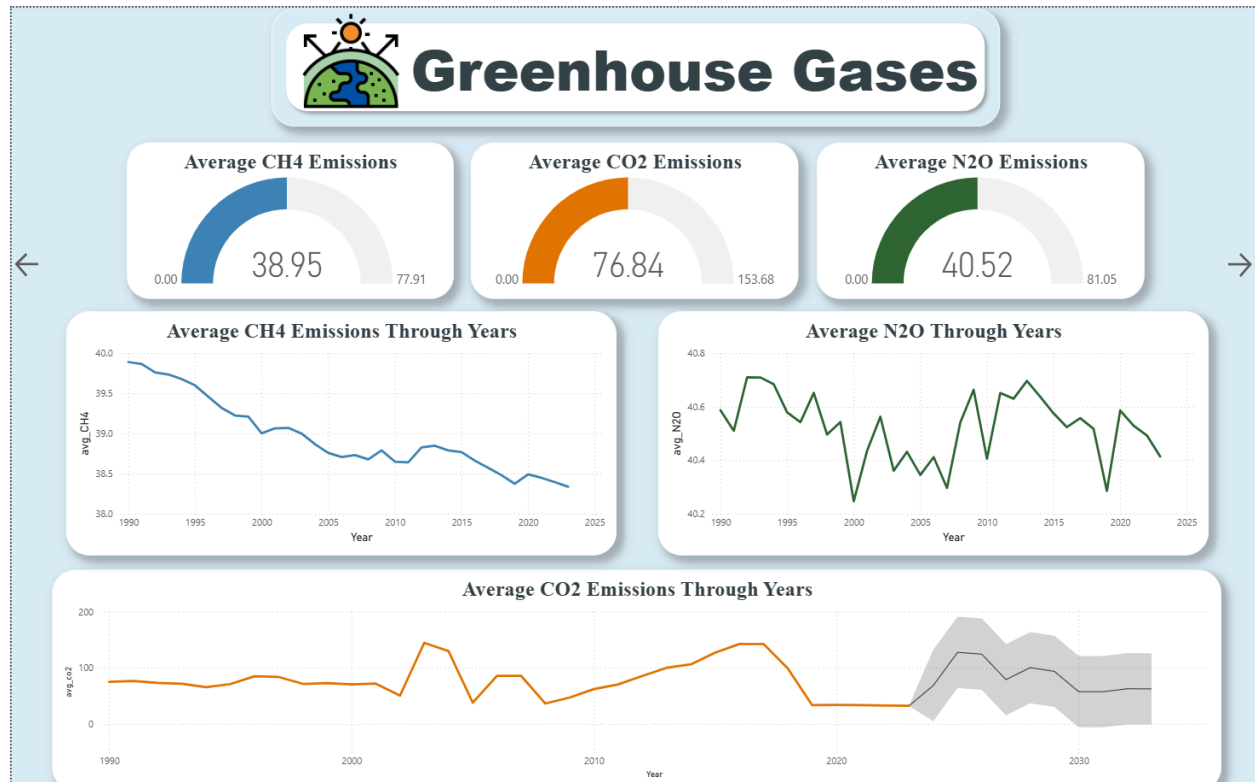
- **Description:** The primary interface mapping global temperature trajectories alongside macroenvironmental indicators.
- **Core Functions:** Multi-region geographical slicers, time-range selectors, and high-level card visualizations for global averages (15.32°C temperature).



Screen 2: Regional Temperature Hotspots

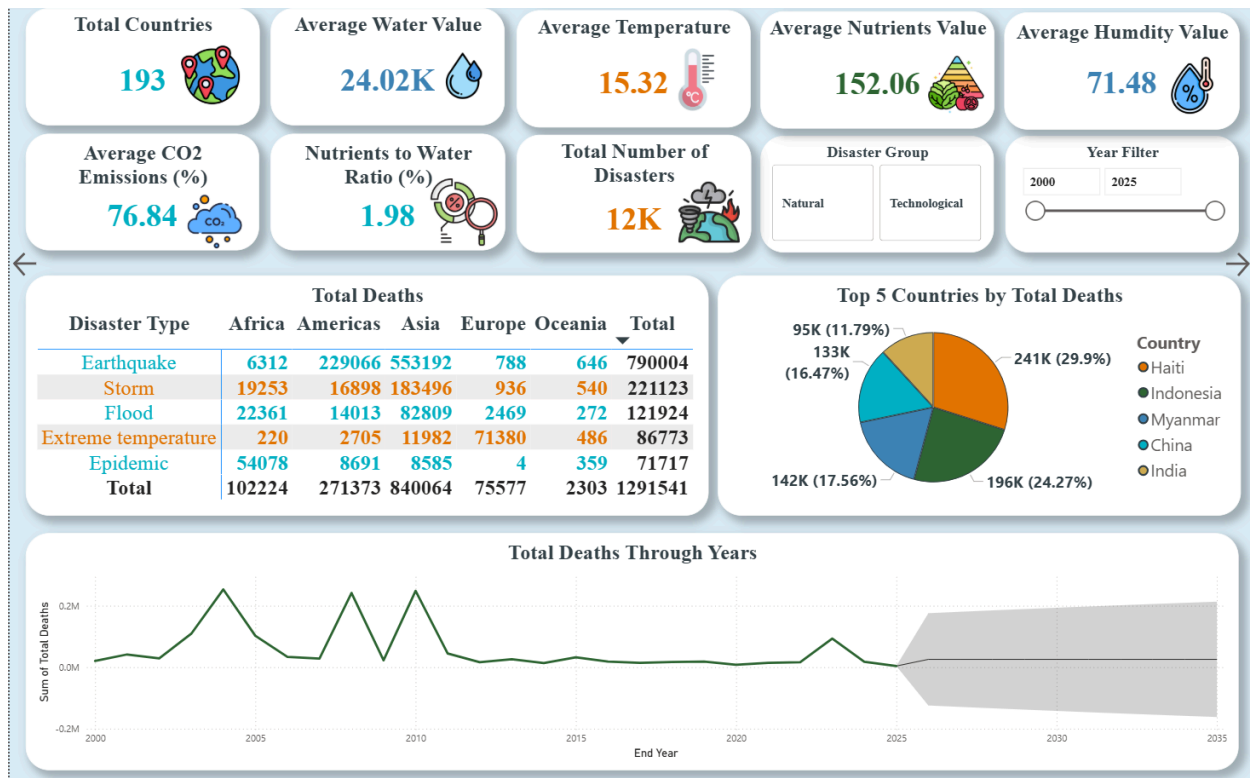
- **Description:** Granular view mapping municipal and regional thermal anomalies across vulnerable urban centers.
- **Core Functions:** City-level bar graphs, sorting dropdowns for peak anomaly identification, and geographical drill-downs for Africa, South Asia, and the Middle East.

Screen 3: Atmospheric Emissions & Greenhouse Gases



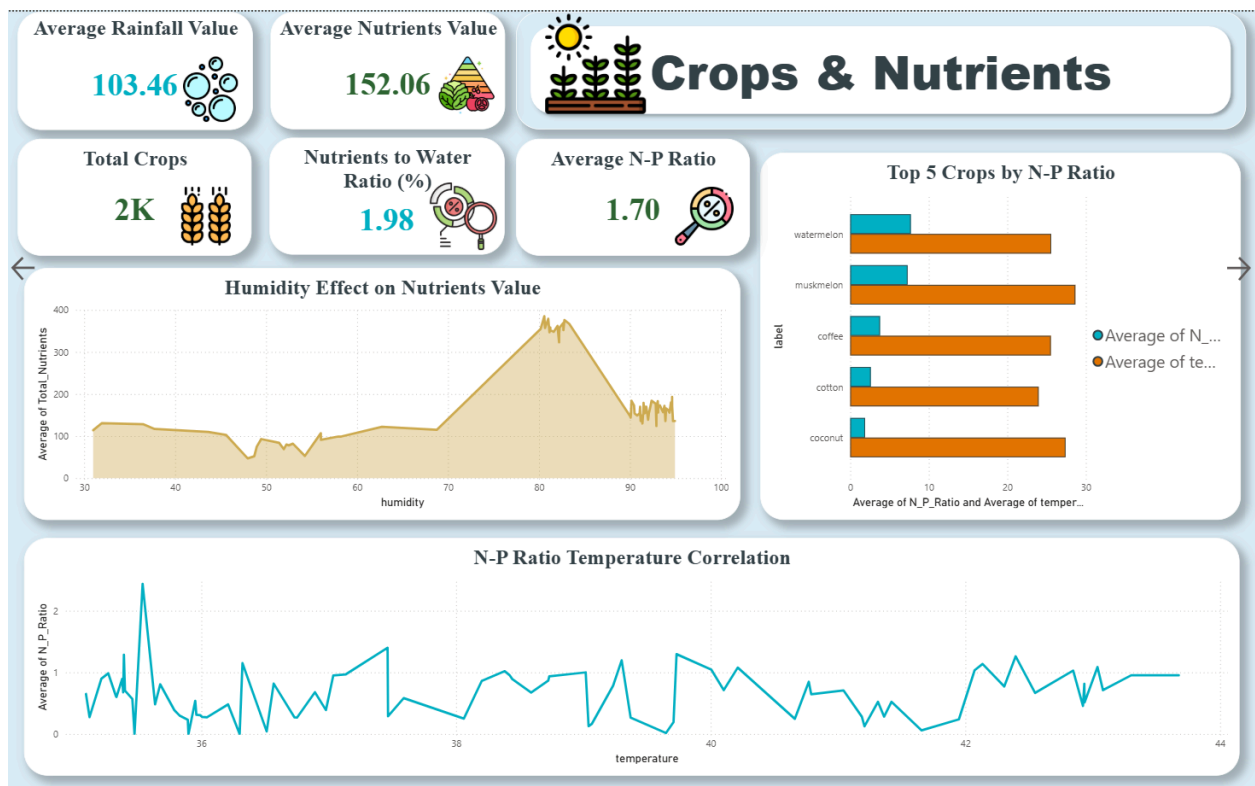
- **Description:** A comparative trend screen tracking atmospheric gas indicators (CO₂, CH₄, N₂O).
- **Core Functions:** Gas element toggles, line charts plotting historical industrialization spikes, and quantile contribution bins (Bronze to Platinum).

Screen 4: Natural Disasters & Fatalities Impact



- **Description:** Spatiotemporal analysis dashboard evaluating disaster frequencies and total human impact over time.
- **Core Functions:** Disaster type selectors (Earthquake, Storm, Flood), interactive heatmaps, and top affected country fatality charts.

Screen 5: Agriculture, Rainfall & Nutrient Relationships



- **Description:** Agricultural analysis dashboard exploring the correlations between crop choices, soil health, and weather variables.
- **Core Functions:** Crop type cross-filtering, scatter plots mapping humidity against total nutrient absorption, and N:P ratio tables.

6. Challenges & Engineering Solutions

Identified Engineering Challenge	Implemented Technical Solution
High Volume of Missing Values & Disparate Formats: Missing entries across multiple columns prevented accurate correlation mapping and dashboard aggregations.	Developed an automated Python looping pipeline that detected data types to apply tailored statistical imputation: column means for numerical attributes and modes for categorical fields.
Data Type Mismatches & Text Whitespace Issues: Text spacing around area names broke data lookups, and numeric indicators were corrupted by nonnumeric strings.	Applied errors='coerce' inside automated Pandas data casting routines to cleanly handle malformed records, and used string manipulation functions to strip trailing spaces across all headers.
Extreme Gaps and Outliers in Resource Metrics: Extreme resource variances between nations skewed baseline charts, hiding details for water-stressed areas.	Implemented IQR-driven statistical outlier detection models and utilized logarithmic scale spacing inside Seaborn visualization frameworks to clearly contrast high and low resource volumes.
Cross-Indicator Relational Complexity: Multi-source datasets from different timelines and regions complicated interactive query performance.	Designed a robust Star Schema model using 1-to-many relationships tied directly to a centralized Global Temperature chronological fact hub.

7. Future Enhancements Roadmap

To scale ClimateScope into an industrial environmental intelligence application, the following enhancement phases have been established:

1. **ML Predictive Forecasting:** Integrate advanced time-series forecasting models (ARIMA and Facebook Prophet) to project regional temperature variations and greenhouse gas emissions 20 years into the future.
2. **Advanced Geospatial Mapping:** Replace static graphs with interactive choropleth and bubble maps utilizing Plotly or Folium libraries to enable dynamic geographic exploration.
3. **Real-Time Live Data Feeds:** Establish automated API pipeline integrations connecting directly to live climate observation streaming networks (NASA GISS, NOAA, and Open-Meteo).
4. **Climate Threshold Alert System:** Build an automated notification service to trigger system alerts when key regional indicators cross hazardous environmental tipping points.
5. **Mobile Layout Optimization:** Refactor Streamlit layout elements using custom CSS frameworks to ensure smooth, responsive performance on mobile devices.
6. **Automated PDF/Excel Export:** Develop localized report generation engines allowing users to dynamically export tailored Excel sheets or PDF summaries based on active dashboard filter states.
7. **Bilingual UI Support (Arabic + English):** Integrate a comprehensive bilingual translation layer across the Streamlit UI to better serve regional stakeholders and Egyptian audience communities.
8. **REST API Service Layer:** Expose processed data models via a secure, high-performance FastAPI service layer to allow open-source access for external academic and research groups.
9. **NLP Insight Generation:** Incorporate natural language processing engines to generate real-time text summaries explaining regional climate trend anomalies automatically.

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